

**Database System Engineering and Distributed Backend
Development**
Course Code: 25CS1302E
Week-2

Create database

```
mysql> CREATE TABLE Books (  
->     book_id INT PRIMARY KEY,  
->     title VARCHAR(100) NOT NULL,  
->     isbn VARCHAR(20) UNIQUE,  
->     published_year INT CHECK (published_year < 2027)  
-> );  
Query OK, 0 rows affected (0.08 sec)  
  
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES  
-> (1, 'The Great Gatsby', '9780743273565', 1925),  
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),  
-> (3, '1984', '9780451524935', 1949);  
Query OK, 3 rows affected (0.02 sec)  
Records: 3  Duplicates: 0  Warnings: 0
```

Create Books Table

```
mysql> CREATE TABLE Books (  
->     book_id INT PRIMARY KEY,  
->     title VARCHAR(100) NOT NULL,  
->     isbn VARCHAR(20) UNIQUE,  
->     published_year INT CHECK (published_year < 2027)  
-> );  
Query OK, 0 rows affected (0.08 sec)  
  
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES  
-> (1, 'The Great Gatsby', '9780743273565', 1925),  
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),  
-> (3, '1984', '9780451524935', 1949);  
Query OK, 3 rows affected (0.02 sec)  
Records: 3  Duplicates: 0  Warnings: 0
```

Create Members table

```
mysql> SELECT * FROM Members;  
+-----+-----+-----+  
| member_id | full_name      | email                      |  
+-----+-----+-----+  
|      101 | John Smith     | john.smith@email.com      |  
|      102 | Emma Wilson    | emma.wilson@email.com     |  
|      103 | Michael Brown  | michael.brown@email.com   |  
+-----+-----+-----+  
3 rows in set (0.00 sec)
```

Create Loans table

```
mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 102 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-03-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

Loans table output

```
mysql> SELECT
  -> m.full_name AS Member_Name,
  -> b.title AS Book_Title
  -> FROM Loans l
  -> INNER JOIN Members m
  -> ON l.member_id=m.member_id
  -> INNER JOIN Books b
  -> ON l.book_id=b.book_id;
+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| John Smith | The Great Gatsby |
| John Smith | To Kill a Mockingbird |
| John Smith | 1984 |
| John Smith | The Great Gatsby |
| Emma Wilson | To Kill a Mockingbird |
| Emma Wilson | The Great Gatsby |
| Emma Wilson | 1984 |
| Michael Brown | 1984 |
| Michael Brown | To Kill a Mockingbird |
| Michael Brown | The Great Gatsby |
+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SELECT
-> published_year,
-> COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1

3 rows in set (0.01 sec)

```
mysql> CREATE TABLE Donation_History (
-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY(book_id)
-> REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.08 sec)

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO Books
-> (book_id,title,isbn,published_year)
-> VALUES
-> (4,'Animal Farm','9780451526342',1945);
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql> INSERT INTO Donation_History
-> (donation_id,book_id,donor_name,donation_date)
-> VALUES
-> (1,4,'Raj Kumar',CURDATE());
Query OK, 1 row affected (0.01 sec)
```

```
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
4	Animal Farm	9780451526342	1945

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Donation_History;
```

donation_id	book_id	donor_name	donation_date
1	4	Raj Kumar	2026-07-30

```
1 row in set (0.00 sec)
```

```
mysql> CREATE INDEX idx_books_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT *  
      -> FROM Books  
      -> WHERE isbn='9780451524935';
```

book_id	title	isbn	published_year
3	1984	9780451524935	1949

```
1 row in set (0.00 sec)
```

```
mysql>
```